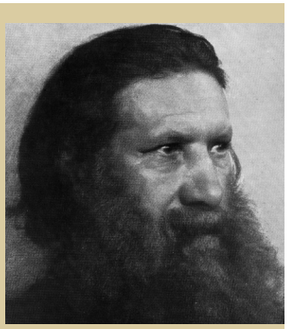
the 'waterframe'. This spinning frame used water power in the place of manpower. The 'waterframe' was too big to be run at home. Thus was born the factory. In 1779 Samuel Crompton invented his 'spinning mule' which included a combination of both the 'spinning jenny' and the 'water frame'. It spun hundreds of threads simultaneously and produced eight fine and coarse threads. Eli Whitney invented the cotton gin in 1793, removing the seed from the cotton.



John Kay



James Hargreaves

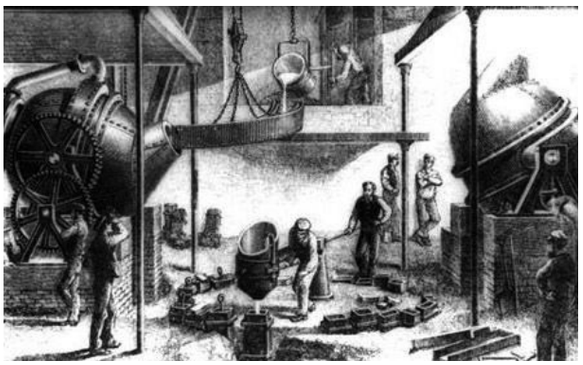
Textile manufacture was at the heart of the Industrial Revolution. Over a span of fifty years, the textile manufacturing industry in Britain witnessed a transformation in the method of production from handmade to machine-made goods. The newly invented machines enabled factories to produce textile goods in large quantities. Derbyshire, Lancashire, Cheshire, Staffordshire, Nottinghamshire, and Yorkshire became the major factory centres. The most notable was Manchester which had more than 50 mills in 1802. These factories involved in mass production were organized on the principle of division of labour.



Lancashire

Iron and Steel

The rolling mill (machine for rolling steel or any other metal into sheets) proved to be fifteen times faster than hammering wrought iron. Hot blast greatly increased fuel efficiency in iron production. In 1856, Henry Bessemer discovered a faster and cheaper method of producing steel. In course of time, iron and steel came to be used in making all machines and in all industries.



Iron and steel industry

Mining

The development of factories by Arkwright and the improvement of the steam engine by James Watt further increased demand for coal. As a result, coal mines became deeper and deeper, making it more and more dangerous. As miners used oil lamps in the mines the risk of explosion was high leading to the death of miners. This was reduced by the invention of a safety lamp by Sir Humphrey Davy in 1815.

The coal production in England increased from 4.7 million tonnes in 1750 to 250 million tonnes in 1900.



Coal mines

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